

KnockKnock

Low-Cost IoT Intrusion Detection for Doors and Windows

IoT: Algorithms and Services

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Project Overview

1 Overview

Mission

Detect intrusion attempts by observing abnormal door and window vibrations together with related physical signals.

- **Wireless:** battery-powered nodes with low-impact installation.
- **Adaptive:** machine learning learns the normal environmental baseline.
- **Actionable:** the system raises alerts and acts in the environment.

Detection Flow

1. Sense
2. Classify
3. Alert
4. Respond





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Problem and Motivation

2 Introduction

- **Risk:** residential and small-building entrances are vulnerable to break-in attempts.
- **Limits:** wired systems can be invasive, expensive, and difficult to deploy in historical or hard-to-reach locations.
- **Vision:** KnockKnock targets a low-cost and scalable alternative with wireless sensors and easy installation.
- **Challenge:** reduce false positives while preserving rapid detection.



Why IoT and What We Measure

2 Introduction

- **Sensing:** nodes collect local vibration/movement signals near doors and windows.
- **Data:** events are aggregated and forwarded to a central backend for storage, analysis, and visualization.
- **Communication** needs:
 - peer-to-peer messaging for fast neighborhood coordination during alarms
 - publish/subscribe uplink for global monitoring and event history
- **Dashboard:** a web interface provides homeowners with a clear and immediate system status view.



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System Architecture and Response

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- **Edge node behavior:**
 - idle low-power mode while waiting for passive trigger events
 - local anomaly scoring based on learned environmental baseline
 - alarm escalation only when confidence exceeds threshold
- **Actions** after alarm confirmation:
 - user notification through web dashboard
 - activation of sirens, floodlights, and optional deadbolt logic
 - neighborhood node wake-up for richer contextual sensing



Constraints and Design Trade-Offs

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- **Bandwidth:**
 - support both local alarm propagation and centralized telemetry
- **Energy:**
 - continuous sensing readiness with strict battery constraints
 - exploration of indoor solar harvesting feasibility
- **Maintenance:**
 - no wiring, rapid installation, and modular mesh expansion
 - operation in hard-to-reach points like high windows or skylights
- **Security (secondary):**
 - device identity, secure transport, and resilience against spoofed alerts



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Next Steps and Closing

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- **Finalize** sensor stack and baseline anomaly model on representative apartment data.
- **Benchmark** network behavior under alarm load (latency, packet loss, battery impact).
- **Validate** actuator policy to balance deterrence effectiveness and false alarm costs.
- **Assess** security hardening and pilot deployment readiness.

KnockKnock

Smart, modular, and practical home intrusion detection.



KnockKnock *Thank you for listening!*

Any questions?